

Solution

- (a) Taking multiple reference points, we get that the BLR emission lags by about **20-25 days**.

Answers from 15 to 25 days are acceptable.

- (b) As the AGN is located very far, the time lag can be approximated as the time taken by UV emission to reach BLR region. Thus,

$$\begin{aligned} R_{\text{BLR}} &= c\Delta t = 3 \times 10^8 \times 20 \times 86\,400 \\ &= 5.2 \times 10^{14} \text{ m} = (0.017 \pm 0.004) \text{ pc} \end{aligned}$$

- (c) As this AGN is 100 Mpc away from us,

$$\begin{aligned} \theta_{\text{BLR}} &= \frac{0.017}{100e6} \times 206265 \\ &= (3.5 \pm 0.9) \times 10^{-5} \text{ arcsec} \end{aligned}$$

- (d) The FWHM is approximately $(85 \pm 5) \text{ \AA}$ and the peak is approximately at $(4940 \pm 5) \text{ \AA}$

$$\begin{aligned} v_{\sigma} &= \frac{\sigma c}{\lambda_{\text{peak}}} = \frac{\text{FWHM} \times c}{2.35 \lambda_{\text{peak}}} = \frac{85 \times 3 \times 10^5}{2.35 \times 4940} \\ v_{\sigma} &= (2200 \pm 140) \text{ km s}^{-1} \end{aligned}$$

- (e)

$$\begin{aligned} M_{\text{vir,BH}} &= \frac{v_{\sigma}^2 R_{\text{BLR}}}{G} = \frac{(2.2 \times 10^6)^2 \times 5.2 \times 10^{14}}{6.674 \times 10^{-11}} \\ &= 3.8 \times 10^{37} \text{ kg} \\ M_{\text{vir,BH}} &= (1.9 \pm 0.7) \times 10^7 M_{\odot} \end{aligned}$$